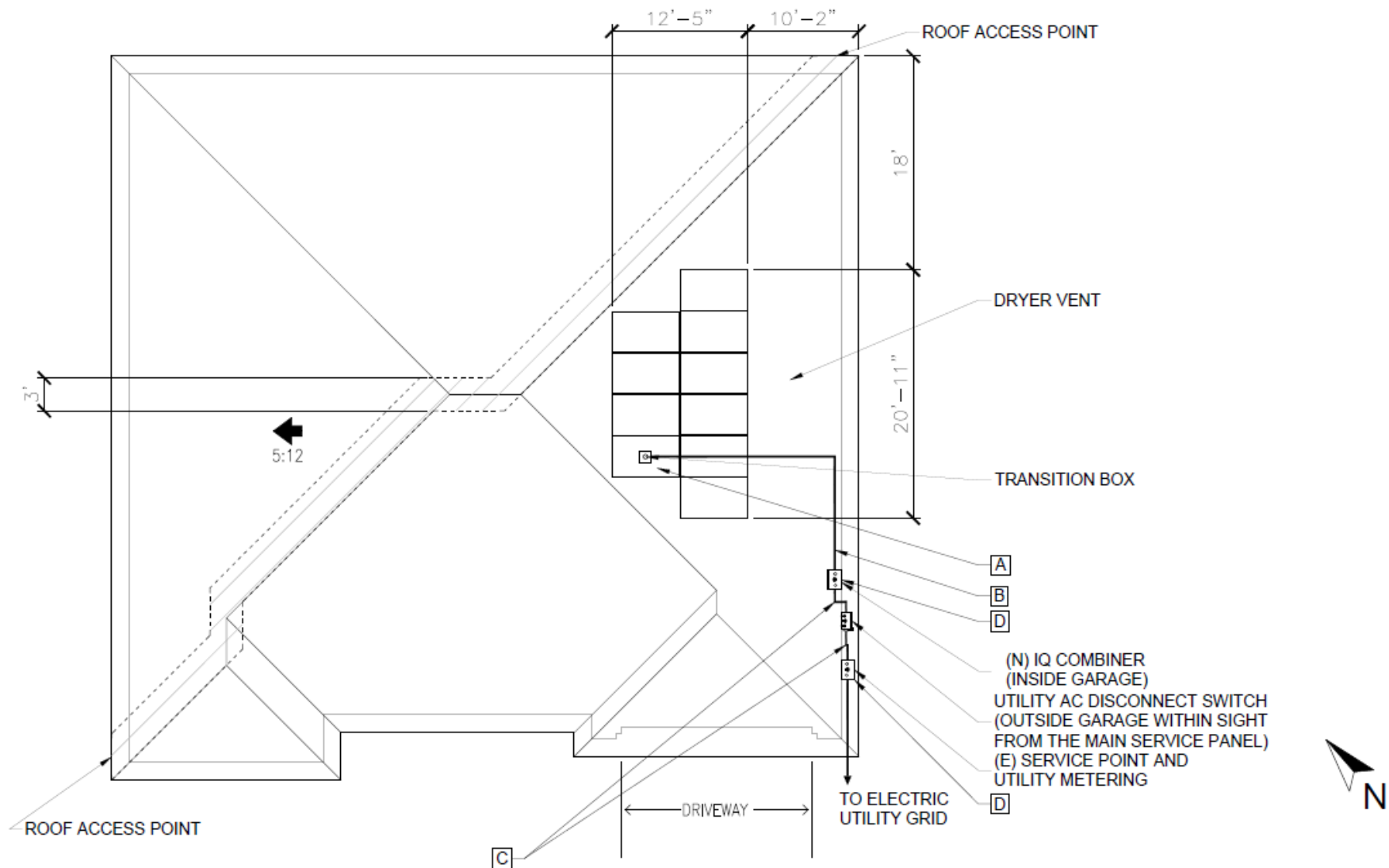
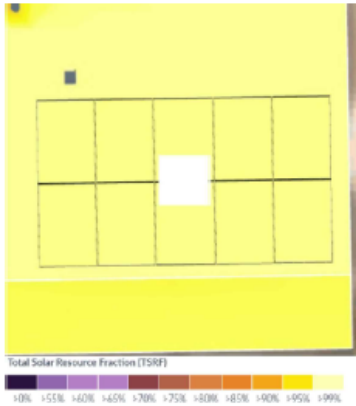


Roof Layout / Array Layout



Installation Array Layout

IRRADIANCE MAP



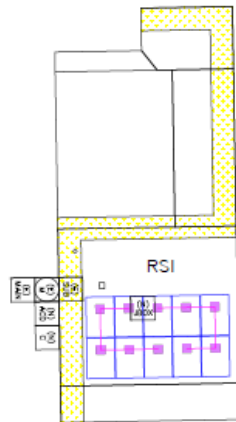
ROOF SECTION	PV MODULE QTY	AZIMUTH	PITCH	TSRF
RS1	10	179	18	95%

FIRE SETBACK



SCALE: 1/16" = 1'-0"

NEW EQUIPMENT LOCATION WALL PHOTO



FRONT OF HOME

SITE ARRAY / EQUIPMENT LOCATION



INTERCONNECTION LOAD CENTER PHOTO



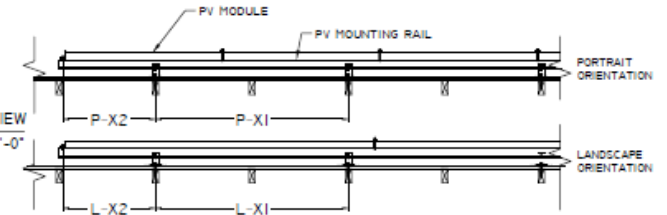
Structural Design Reference

STRUCTURAL INSTALLATION SCHEDULE AND CRITERIA

ROOF MEMBRANE		SPAN AREA		TAG	SPAN
ROOF TYPE:	COMPOSITION SHINGLE	RAIL - PORTRAIT - MODULE ORIENTATION			
ROOF SHEATHING TYPE:	7/16" OSB	X- SPACING	P-X1	48 IN. O.C. MAX.	
		X-CANTILEVER	P-X2	23 IN. MAX.	
		Y- SPACING	P-Y1	40.2 IN. MIN. - 55.9 IN. MAX.	
		Y-CANTILEVER	P-Y2	6 IN. MIN. - 13.8 IN. MAX.	
ARRAY PARAMETERS		RAIL - LANDSCAPE - MODULE ORIENTATION			
TOTAL ROOF AREA (SQ. FT.)	1284	X- SPACING	L-X1	48 IN. O.C. MAX.	
TOTAL PV MODULE AREA (SQ. FT.)	210	X-CANTILEVER	L-X2	23 IN. MAX.	
% PV MODULE ROOF COVERAGE	16.36%	Y- SPACING	L-Y1	24.9 IN. MIN. - 28.9 IN. MAX.	
		Y-CANTILEVER	L-Y2	7.9 IN. MIN. - 9.8 IN. MAX.	
PV RACKING					
RACKING:	UNIRAC NXT				
RACKING TYPE:	RAIL UNIRAC STRONGHOLD BUTYL ATT KIT #145				
STANDOFF:	MILL				
STANDOFF TYPE:	L-FOOT & BUTYL TAPE HASTIC				
FASTENER:	3" #14-14 AB SCREW				
STRUCTURAL FRAMING					
ROOF SECTION	STRUCTURE TYPE	RAFTER / TC SIZE (IN)	RAFTER / TC SPACING (IN)	STRUCTURAL UPGRADE	UPGRADE SIZE & LENGTH (FT)
RSI	MANUFACTURED WOOD TRUSS	2x4	24	NONE - SINGLE PLY (1X)	NONE

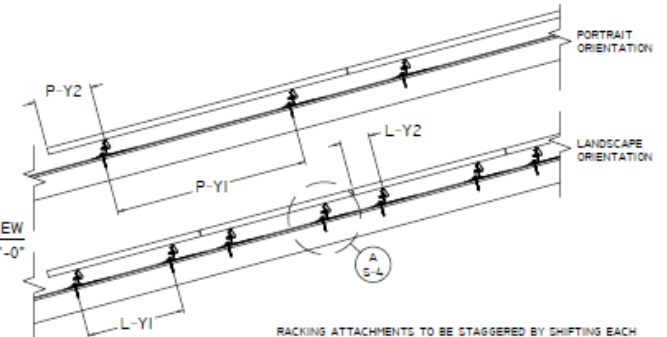
PV ARRAY DETAIL, FRONT VIEW

SCALE: 3/8" = 1'-0"



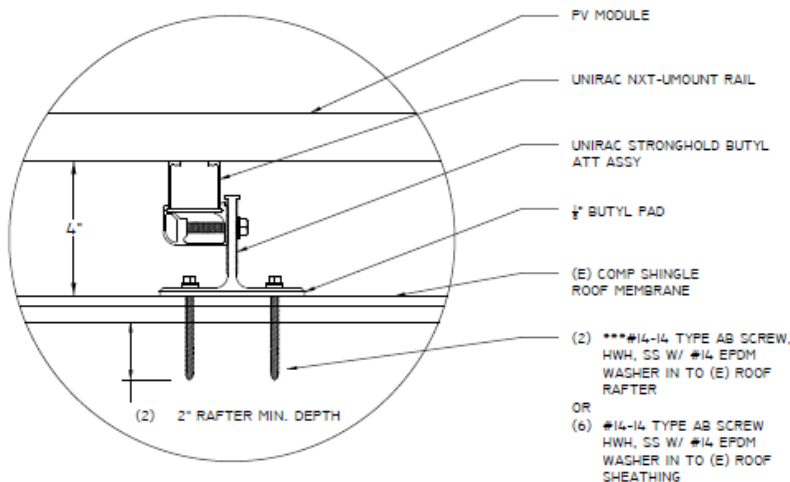
PV ARRAY DETAIL, SIDE VIEW

SCALE: 3/8" = 1'-0"



RACKING ATTACHMENTS TO BE STAGGERED BY SHIFTING EACH SUBSEQUENT ROW OF ATTACHMENTS ONE RAFTER OVER TO DISTRIBUTE LOAD ACROSS ALL FRAMING MEMBERS UNDER PV ARRAY.

*** ONLY 2 FASTENERS REQUIRED IN TO (E) ROOF RAFTER/TOP CHORD. IF DURING INSTALLATION THE (E) RAFTER/TOP CHORD IS MISSED, CONTINUE TO DRIVE FASTENER UNTIL 2 LAND IN TO THE (E) RAFTER/TOP CHORD. IF ALL MISS, (6) IN (E) SHEATHING ONLY ARE SUFFICIENT AS LONG AS ATTACHMENTS SPAN NO GREATER THAN 24" OC.



A COMP SHINGLE - STRONGHOLD BUTYL ATT KIT - STANDOFF DETAIL
SCALE: 3" = 1'-0"

Structural Design Reference

PV SYSTEM STRUCTURAL SPECIFICATIONS AND CALCULATIONS

DESIGN LOCATION AND SITE SPECIFICATIONS

JURISDICTION	TOWN OF BERTHOUD
STATE	COLORADO
ADOPTED LOAD STANDARD	ASCE 7-16
OCCUPANCY / RISK CATEGORY	II
BASIC WIND SPEED (10-M (3-SEC GUST))	125
WIND EXPOSURE CATEGORY	B
GROUND SNOW LOAD (PSF) (Pa)	30
BASE ELEVATION (FT)	5031
STRUCTURE TYPE	INHABITED - DABLE / FLAT ROOF
MIN. ROOF SLOPE (DEG.)	16
MEAN ROOF HEIGHT (FT.)	20

RACKING AND ATTACHMENT SPECIFICATIONS

MODULE MANUFACTURER / TYPE	SILFAR SOLAR SIL-440 60
SOLAR MODULE HEIGHT (IN.)	49.3
SOLAR MODULE LENGTH (IN.)	67.9
SOLAR MODULE WIDTH (IN.)	44.6
SOLAR MODULE AREA (SQ. FT)	21
PV RACKING	UNIRAC NIT
PV RACKING TYPE	RAIL
PV ROOF ATTACHMENT	UNIRAC STRONGHOLD BUTYL ATT KIT #4S MILL
PV ROOF ATTACHMENT FASTENER	(2) 3" #14-14 AB SCREW(S)
MANUFACTURER UPLIFT CAPACITY (LBS)	779
MANUFACTURER LATERAL CAPACITY (LBS)	320
PORTRAIT ATT. SPACING (IN. O.C.)	48
LANDSCAPE ATT. SPACING (IN. O.C.)	48
MAX PORTRAIT RAIL CANTILEVER SPACING (IN. O.C.)	23
MAX LANDSCAPE RAIL CANTILEVER SPACING (IN. O.C.)	23
# OF ATTACHMENT POINTS	24
RACKING DEAD LOAD (PSF)	0.6
SOLAR MODULE DEAD LOAD (PSF)	2.20
TOTAL PV ARRAY DEAD LOAD (PSF)	3.00
MAX. ROOF POINT DEAD LOAD (PSF)	34.0

DESIGNED VELOCITY PRESSURE CALCULATIONS

VELOCITY PRESSURE (PSF) = $0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	ASCE 7-16 (C6.5)
TERRAIN EXPO. CONSTANT (K ₁)	7.0
TERRAIN EXPO. CONSTANT (Z _s)(FT)	1200
VF EXPOSURE COEFFICIENT (K ₂)	0.70
TOPOGRAHIC FACTOR (K ₃)	1.0
WIND DIRECTIONALITY FACTOR (K ₄)	0.9
GROUND ELEVATION FACTOR (K ₅)	1.0
Q _H (PSF)	25.0

DESIGNED ROOF SNOW LOAD CALCULATIONS

SLOPED ROOF SNOW LOAD (PSF)	ASCE 7-16
GROUND SNOW LOAD (PSF) (Pa)	30.0
NO PV - WARM NON SLIPPERY SURFACE	
JURISDICTION MINIMUM ROOF SNOW LOAD (PSF)	30
PV - COLD SLIPPERY SURFACE	
JURISDICTION MINIMUM ROOF SNOW LOAD (PSF)	30

RACKING - ATTACHMENT AND FASTENER CONNECTION ZONE WIND PRESSURE AND POINT LOAD

TABLE 26.10-1	TABLE 26.10-1	TABLE 26.10-1	TABLE 26.10-1	TABLE 26.10-1	TABLE 26.10-1
DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$	DESIGN WIND PRESSURE (PSF) = $W = 0.00256(K_1)(K_2)(K_3)(K_4)(V^2)$
ARRAY EDGE FACTOR (K _e)	1	1	1	1	1
	UPLIFT	COMPRESSION	UPLIFT	COMPRESSION	UPLIFT
	ZONE 1	ZONE 2A	ZONE 2B	ALL ZONES	ALL ZONES
RAIL - PORTRAIT MODULE ORIENTATION	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.
SOLAR PANEL PRESSURE CO. FACTOR (C _f)	0.90	0.90	0.90	0.90	0.90
EXTERNAL PRESSURE COEFF. (C _e)	-2.0	-3.0	-3.0	0.5	-3.0
ASD PRESSURE (W)(PSF)	-59.93	-59.99	-71.67	10.58	22.21
PV DEAD LOAD (D)	3.00	3.00	3.00	3.00	0.00
SNOW OR LIVE LOAD (S OR L)(PSF)	0.00	0.00	0.00	30.00	20.00
MAX. TRIBUTARY AREA (SQ. FT)	11.3	11.3	8.3	11.3	11.3
GOV. LOAD COMBINATION (2.4.1)	0.60+0.6W	0.60+0.6W	0.60+0.6W	0+0.45W+0.75S	0+0.75L+0.45W
MAX. LOAD (LBS)	-220.4	-253.7	-350.2	320.9	127.9
RAIL - LANDSCAPE MODULE ORIENTATION	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.	48 IN. O.C.
SOLAR PANEL PRESSURE CO. FACTOR (C _f)	0.90	0.90	0.90	0.90	0.90
EXTERNAL PRESSURE COEFF. (C _e)	-2.0	-3.0	-3.0	0.7	-3.0
ASD PRESSURE (W)(PSF)	-59.93	-59.99	-71.67	15.98	22.21
PV DEAD LOAD (D)	3.00	3.00	3.00	3.00	0.00
SNOW OR LIVE LOAD (S OR L)(PSF)	0.00	0.00	0.00	30.00	20.00
TRIBUTARY AREA (SQ. FT)	7.43	7.43	5.58	7.43	7.43
GOV. LOAD COMBINATION (2.4.1)	0.60+0.6W	0.60+0.6W	0.60+0.6W	0+0.45W+0.75S	0+0.75L+0.45W
MAX. LOAD (LBS)	-164.7	-253.7	-230.4	227.1	99.3

RACKING - ATTACHMENT AND FASTENER CAPACITY / DEMAND CHECK

UNIRAC STRONGHOLD BUTYL ATT KIT #4S MILL - (2) 3" #14-14 AB SCREW(S)	NO. 12.2	
LAB SCREW WITHDRAWAL DESIGN VALUE (LBS) = $W = 1600(0.3/2)(0.3/4)$	12.2.1	
ROOF ATTACHMENT FASTENER (D) (IN.)	6/25	TABLE 2.3.1
FASTENER QTY PER ATTACHMENT	2	TABLE 2.3.1
FASTENER EMBEDMENT DEPTH (IN.)	2.0	
LUGGER SPECIFIC GRAVITY (G)	0.5	
LOAD DURATION FACTOR (Cd)	1.6	
DRYING COEFFICIENT	1.4	
WITHDRAWAL DESIGN VALUE(W)(LBS) / (IN.)	219.2	
LAB SCREW WITHDRAWAL CAPACITY (LBS)	997.6	
	UPLIFT	LATERAL
MANUFACTURER MAX. CAPACITY (LBS)	779.0	320.0
MAX. LAB SCREW CAPACITY (LBS)	997.6	
MAX. ATT. WITHDRAWAL DEMAND (LBS)	325.7	126.0
	OK	OK

RACKING - RAIL, SPAN AND CANTILEVER CHECK - ALL ZONES

UNIRAC NIT	MAX. HORIZONTAL RAIL SPAN (L)(FT.)	4.0
	MAX. VERTICAL SPACING BETWEEN RAILS (FT.)	4.7
	MAX. RAIL CANTILEVER SPAN (L)(FT.)	1.9
	GOV. LOAD COMBINATION (2.4.1)	0+0.45W+0.75S
	TOTAL LOAD (PSF)	36.7
	TOTAL LINEAR LOAD (W)(LBS. / FT.)	99.7
RAIL - SPAN CHECK	ALLOWABLE MANU. BENDING MOMENT (LBS. / FT.)	223
	ACTUAL MAX. BENDING MOMENT (LBS. / FT.)	197.4
	ALLOWABLE DEFLECTION (IN.)	0.80
	ACTUAL MAX. DEFLECTION (IN.)	0.16
		OK
RAIL - CANTILEVER CHECK	OVERHANG	
	ALLOWABLE MANU. BENDING MOMENT (LBS. / FT.)	223
	ACTUAL MAX. BENDING MOMENT (LBS. / FT.)	181.3
	ALLOWABLE MANU. DEFLECTION (LBS)(IN.)	0.58
	ACTUAL MAX. DEFLECTION (IN.)	0.04
SPAN REACTION	ALLOWABLE MANU. BENDING MOMENT (LBS. / FT.)	331
	ACTUAL MAX. BENDING MOMENT (LBS. / FT.)	117.2
	ALLOWABLE MANU. DEFLECTION (LBS)(IN.)	0.80
	ACTUAL MAX. DEFLECTION (IN.)	0.01
ATTACHMENT / CONNECTION REACTION	UPLIFT CAPACITY (LBS)	779.0
	ACTUAL MAX. REACTION UPLIFT FORCE (LBS)	431.9
		OK

RACKING - MODULE CLAMP CAPACITY / DEMAND CHECK

710 CLAMP: UNIRAC NIT UNJOINT COMBO CLAMP - DARK			
510 CLAMP: UNIRAC NIT UNJOINT COMBO CLAMP - DARK			
	UPLIFT	SLIDING	LATERAL
MANUFACTURER MAX. CAPACITY (LBS)	1620.0	851.0	399.0
MAX. CLAMP WITHDRAWAL DEMAND (LBS)	325.7	126	126
	OK	OK	OK

Structural Design Reference

TABLE 1: ATTACHMENT SPACING
BASIC WIND SPEED 14-110 MPH
RISK CATEGORY I
EXPOSURE C
GROUND SNOW LOAD: 14 PSF

	MAX SPACING	MAX RAIL CANTILEVER
ZONE 1	72 IN	24 IN
ZONE 2	72 IN	24 IN

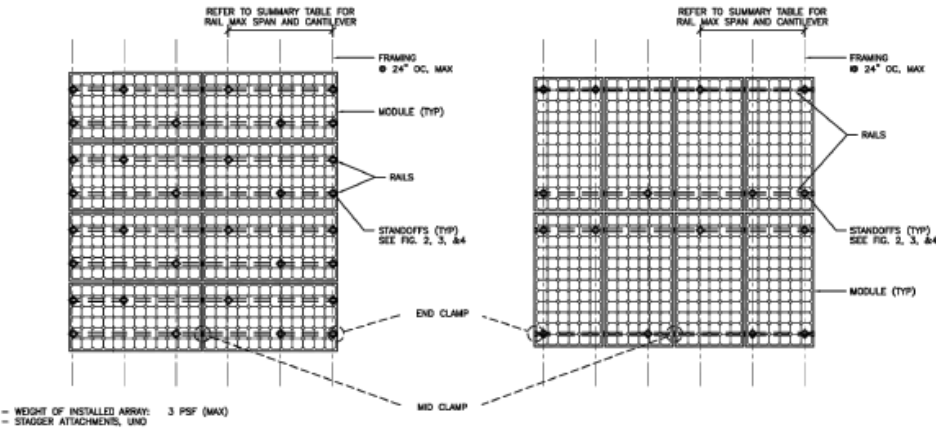


FIGURE 1: PLAN VIEW, TYPICAL ARRAY LAYOUT

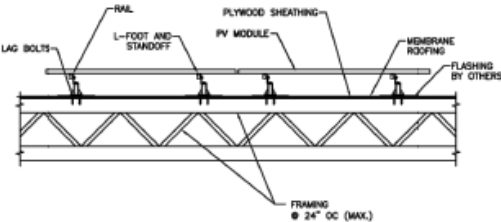


FIGURE 2: SIDE VIEW DETAIL

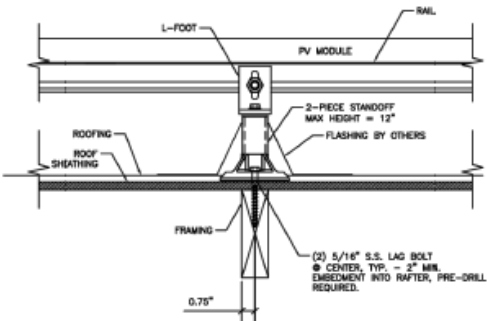


FIGURE 3: ROOF ATTACHMENT DETAIL

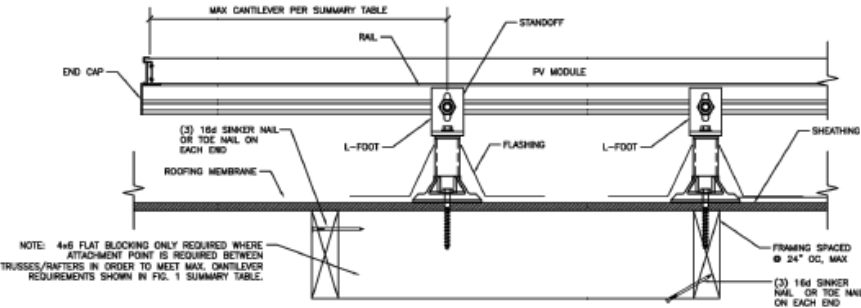


FIGURE 4: ROOF ATTACHMENT DETAIL @ BLOCKED LOCATIONS

Electrical Design Reference

AC BRANCH CIRCUIT 1	#10
NUMBER OF MODULES PER BRANCH CIRCUIT	10
VOLTAGE	240 V
BRANCH CURRENT: $I_{OUT} =$	13.30 A
MINIMUM WIRE AMPACITY: $I_{MAX} = I_{OUT} \times 1.25$	16.63 A
CONDUCTOR DE-RATING	
MAXIMUM AMBIENT TEMPERATURE	47°C
TEMPERATURE ADDER	22°C
TEMPERATURE USED FOR AMPACITY DE-RATING	69°C
TEMPERATURE DE-RATING COEFFICIENT	0.58
FILL DE-RATING COEFFICIENT	1.0
$I_{WIRE_MIN} = I_{OUT} / TEMP_COEFF / FILL_COEFF$	22.93 A
WIRE SIZE SELECTED BASED ON AMPACITY	#10
WIRE SIZE AMPACITY	30A
CONDUCTOR SIZE PER NEC	#10
CONDUCTOR SIZE ADJUSTED FOR VOLTAGE DROP	#10
VOLTAGE DROP AT 70 FT DISTANCE	0.96%
OVERCURRENT PROTECTION	20A, 2P
MINIMUM OCPD = $I_{OUT} \times 1.25$	16.63 A
OCPD SIZE	20 A

AC BRANCH CIRCUIT 2	#10
NUMBER OF MODULES PER BRANCH CIRCUIT	10
VOLTAGE	240 V
BRANCH CURRENT: $I_{OUT} =$	13.30 A
MINIMUM WIRE AMPACITY: $I_{MAX} = I_{AC_OUT} \times 1.25$	16.63 A
CONDUCTOR DE-RATING	
MAXIMUM AMBIENT TEMPERATURE	47°C
TEMPERATURE ADDER	22°C
TEMPERATURE USED FOR AMPACITY DE-RATING	69°C
TEMPERATURE DE-RATING COEFFICIENT	0.58
FILL DE-RATING COEFFICIENT	1.0
$I_{WIRE_MIN} = I_{OUT} / TEMP_COEFF / FILL_COEFF$	22.93 A
WIRE SIZE SELECTED BASED ON AMPACITY	#10
WIRE SIZE AMPACITY	30A
CONDUCTOR SIZE PER NEC	#10
CONDUCTOR SIZE ADJUSTED FOR VOLTAGE DROP	#10
VOLTAGE DROP AT 70 FT DISTANCE	0.96%
OVERCURRENT PROTECTION	20A, 2P
MINIMUM OCPD = $I_{OUT} \times 1.25$	16.63 A
OCPD SIZE	20 A

IQ COMBINER TO POINT OF INTERCONNECTION	#8
VOLTAGE	240 V
SUM OF BRANCHES: $I_{OUT_TOTAL} =$	26.60 A
MINIMUM WIRE AMPACITY: $I_{MAX} = I_{OUT_TOTAL} \times 1.25$	33.25 A
CONDUCTOR DE-RATING	
MAXIMUM AMBIENT TEMPERATURE	47°C
TEMPERATURE USED FOR AMPACITY DE-RATING	47°C
TEMPERATURE DE-RATING COEFFICIENT	0.82
FILL DE-RATING COEFFICIENT	1.0
$I_{WIRE_MIN} = I_{OUT} / TEMP_COEFF / FILL_COEFF$	32.44 A
WIRE SIZE SELECTED BASED ON AMPACITY	#8
WIRE SIZE AMPACITY	40A
CONDUCTOR SIZE PER NEC	#8
CONDUCTOR SIZE ADJUSTED FOR VOLTAGE DROP	#8
VOLTAGE DROP AT 10 FT DISTANCE	0.17%
OVERCURRENT PROTECTION	40A, 2P
MINIMUM OCPD = $I_{OUT} \times 1.25$	33.25 A
OCPD SIZE	40 A

[illegible]

(E) UTILITY METER
120/240V AC, 60HZ,
SINGLE PHASE

(E)1000A MAIN
SERVICE PANEL
(E)150A/2P MAIN
DISCONNECT

SECONDARY POWER
SOURCE (PV OCPD) TO
BE LOCATED AT
OPPOSITE END OF
BUSBAR FROM PRIMARY
SOURCE (MAIN OCPD)
(NEC 705.12)

(E)150A/2P
(E)LOADS

(E)100A/2P
(E)100A SUBPANEL
(N)150A/2P
(E)LOADS
(E)LOADS

(E) GROUNDING
ELECTRODE
SYSTEM TO BE
VERIFIED OR
INSTALLED

UNDERGROUND
SERVICE LATERAL

(N) PV UTILITY AC
DISCONNECT
30A, 240V, 2P
NON- FUSED, LOCKABLE,
VISIBLE OPEN

(N) ENPHASE IQ COMBINER
(PV CIRCUITS ONLY)

INTEGRATED
15A/2P MAX
TO IQ ENVVOY
GATEWAY

(N)100A/2P

(N) JUNCTION BOX
(OPTIONAL - FOR
CONDUCTOR SPLICE)

(N) JUNCTION BOX
(OPTIONAL - FOR
CONDUCTOR SPLICE)

(10) SILFAQ SOLAR SIL-440 QD PV MODULE(S)
UL 1703 LISTED

(10) ENPHASE IQ8MC-72-H-US MICROINVERTER(S) [240V]
UL 1741 LISTED

(1) CIRCUIT
OF 10
MODULES

Electrical Design Reference

PV SYSTEM ELECTRICAL SPECIFICATIONS AND CALCULATIONS

DESIGN LOCATION AND TEMPERATURES

TEMPERATURE DATA SOURCE	ASHRAE
STATE	COLORADO
JURISDICTION	TOWN OF BERTHOUD
WEATHER STATION	FORT COLLINS AWOS
ASHRAE EXTREME LOW TEMP (°C)	-23
ASHRAE 2% HIGH TEMP (°C)	34
DESIGNED MAX. SYSTEM VDROP / VRISE	4.00%
PV MODULE SPECIFICATIONS	
SILFAB SOLAR SIL-440 00	
RATED POWER (PMAX) (W DC)	440
MAXIMUM POWER VOLTAGE (VMP DC)	33.41
MAXIMUM POWER CURRENT (IMP DC)	13.17
OPEN CIRCUIT VOLTAGE (VOC DC)	38.07
SHORT CIRCUIT CURRENT (ISC DC)	14.22
PMP/VMP TEMP. COEFFICIENT	-0.20
VOC TEMP. COEFFICIENT	-0.24
SERIES FUSE RATING (A DC)	25
ADJ. MODULE VOC @ ASHRAE LOW TEMP (°C)	43.5
ADJ. MODULE VMP @ ASHRAE 2% AVG. HIGH TEMP (°C)	29.4

INVERTER(S) SPECIFICATIONS

TYPE	ENPHASE 108MC-72-H-US	--
MAX. OR RECOMMENDED MODULE POWER (W)	440	0
MAXIMUM INPUT DC OPEN-CIRCUIT VOLTAGE (VOC)	60	0
MINIMUM START VOLTAGE (V)	32	0
MAXIMUM DC START VOLTAGE(V)	58	0
MAXIMUM INPUT CURRENT (ISC) (A)	20	0
MAX. CONTINUOUS OUTPUT POWER (VA)	320	0
MAX. CONTINUOUS OUTPUT CURRENT (A)	1.33	0
NOMINAL (L-L) OUTPUT VOLTAGE	240	0
CEC WEIGHTED EFFICIENCY (%)	97.0%	0.0%

SYSTEM ELECTRICAL SPECIFICATIONS

NUMBER OF MODULES PER CIRCUIT	10
DC POWER RATING PER CIRCUIT (STC)(W DC)	4400
TOTAL MODULE QUANTITY	10 PV MODULES
STC DC POWER RATING OF ARRAY	4400W DC
INVERTER OUTPUT CIRCUIT CURRENT(A AC)	13.3
125% INVERTER OUTPUT CIRCUIT CURRENT(A AC)	16.63
CIRCUIT OCPD RATING (A)	20
COMBINED INV. CONT. OUTPUT CURRENT x 125% (A AC)	16.7
PV POWER PRODUCTION SYSTEM OCPD RATING (A)	20
MAX. ARRAY STC-AC POWER (W)	3200
MAX. ARRAY CEC-AC POWER (W)	4027

AC VOLTAGE RISE CALCULATIONS

	DIST (FT)	COND.	VRISE(V)	VEND(V)	%VRISE
VRISE SEC. 1 (MICRO TO JBOX) *	28.8	12 CU.	1.5	241.5	0.64%
VRISE SEC. 2 (JBOX TO COMBINER BOX)	48	12 CU.	2.6	242.6	1.06%
VRISE SEC. 3 (COMBINER BOX TO POI)	16	10 CU.	0.5	240.5	0.21%
TOTAL VRISE			4.6	244.6	1.92% OK

* 8 MICROINVERTER MAX SUB-BRANCH CIRCUIT SIZE TO COMPLY WITH VRISE CALCULATIONS.

RACEWAY / CONDUCTOR CALCULATIONS

MICROINV. TO JUNCTION BOX (1)

MAX INVERTER OUTPUT CIRCUIT CURRENT *	13.3 A AC	2C, TC-ER, CU.
CONDUCTOR SIZE / INSULATION / TYPE *	12 AWG	
CONDUCTOR AMP. RATING @ 90°C *	30 A	
PER NEC 690.8(B)(1)(W/O CORRECTION FACTORS)		
MAX INVERTER OUTPUT CURRENT X125% = 16.6	A AC	
PER NEC 690.8(B)(2)(WITH CORRECTION FACTORS)		
AMB. TEMP. AMP. CORRECTION *	0.94	
# OF CONDUCTORS IN RACEWAY CORRECTION *	1.0	
ADJUSTED CONDUCTOR AMPACITY (A) *	28.8 A AC	
LARGER AMPACITY OF 690.8(B)(1) OR (B)(2) *	16.6 < 28.8	
(B)(1) - W/O CORRECTION FACTORS		
LARGER AMPACITY COMPLIANCE *	30.0 > 16.6	OK
RACEWAY SIZE / TYPE *	1 1/4 IN.	EMT OR NO RACEWAY
CONDUCTOR(S) / CABLE(S) CROSS-SECTION AREA (IN. ²) *	0.311 IN. ²	
CROSS-SECTIONAL AREA OF RACEWAY(IN. ²)	1.406 IN. ²	
% ALLOWABLE RACEWAY FILL (NEC CH. 9, TBL. 1) *	40% > 21%	OK

JUNCTION BOX TO JUNCTION BOX (2)

MAX INVERTER OUTPUT CIRCUIT CURRENT *	13.3 A AC	2C, NM-B W/G, CU.
CONDUCTOR SIZE / INSULATION / TYPE *	12 AWG	
CONDUCTOR AMP. RATING @ 90°C *	20 A	
PER NEC 690.8(B)(1)(W/O CORRECTION FACTORS)		
MAX INVERTER OUTPUT CURRENT X125% = 16.6	A AC	
PER NEC 690.8(B)(2)(WITH CORRECTION FACTORS)		
AMB. TEMP. AMP. CORRECTION *	0.91	
# OF CONDUCTORS IN RACEWAY CORRECTION *	1.0	
ADJUSTED CONDUCTOR AMPACITY (A) *	18.2 A AC	
LARGER AMPACITY OF 690.8(B)(1) OR (B)(2) *	16.6 < 18.2	
(B)(1) - W/O CORRECTION FACTORS		
LARGER AMPACITY COMPLIANCE *	20.0 > 16.6	OK
RACEWAY SIZE / TYPE *	CABLE	

JUNCTION BOX TO COMBINER BOX (3)

MAX INVERTER OUTPUT CIRCUIT CURRENT *	13.3 A AC	THHN / THWN-2, CU.
CONDUCTOR SIZE / INSULATION / TYPE *	12 AWG	
CONDUCTOR AMP. RATING @ 75°C *	20 A	
PER NEC 690.8(B)(1)(W/O CORRECTION FACTORS)		
MAX INVERTER OUTPUT CURRENT X125% = 17.0	A AC	
PER NEC 690.8(B)(2)(WITH CORRECTION FACTORS)		
AMB. TEMP. AMP. CORRECTION *	0.94	
# OF CONDUCTORS IN RACEWAY CORRECTION *	1.0	
ADJUSTED CONDUCTOR AMPACITY (A) *	18.8 A AC	
LARGER AMPACITY OF 690.8(B)(1) OR (B)(2) *	17.0 < 18.8	
(B)(1) - W/O CORRECTION FACTORS		
LARGER AMPACITY COMPLIANCE *	20.0 > 17.0	OK
RACEWAY SIZE / TYPE *	3/4 IN.	EMT
CONDUCTOR(S) / CABLE(S) CROSS-SECTION AREA (IN. ²) *	0.040 IN. ²	
CROSS-SECTIONAL AREA OF RACEWAY(IN. ²)	0.533 IN. ²	
% ALLOWABLE RACEWAY FILL (NEC CH. 9, TBL. 1) *	40% > 7%	OK

COMBINER BOX TO MAIN PV OCPD (10)

COMBINED INVERTER CONTINUOUS OUTPUT CURRENT *	13.3 A AC	THHN / THWN-2, CU.
CONDUCTOR SIZE / INSULATION / TYPE *	10 AWG	
CONDUCTOR AMP. RATING @ 75°C *	35 A	
PER NEC 690.8(B)(1)(W/O CORRECTION FACTORS)		
MAX COMBINED INVERTER CONT. OUTPUT CURRENT X125% = 17.0	A AC	
PER NEC 690.8(B)(2)(WITH CORRECTION FACTORS)		
AMB. TEMP. AMP. CORRECTION *	0.94	
# OF CONDUCTORS IN RACEWAY CORRECTION *	1.0	
ADJUSTED CONDUCTOR AMPACITY (A) *	32.9 A AC	
LARGER AMPACITY OF 690.8(B)(1) OR (B)(2) *	17.0 < 32.9	
(B)(1) - W/O CORRECTION FACTORS		
LARGER AMPACITY COMPLIANCE *	35.0 > 17.0	OK
RACEWAY SIZE / TYPE *	3/4 IN.	EMT
CONDUCTOR(S) / CABLE(S) CROSS-SECTION AREA (IN. ²) *	0.084 IN. ²	
CROSS-SECTIONAL AREA OF RACEWAY(IN. ²)	0.533 IN. ²	
% ALLOWABLE RACEWAY FILL (NEC CH. 9, TBL. 1) *	40% > 16%	OK

Electrical Design Reference

FIGURE 1: ELECTRICAL SIGNAGES/STICKERS

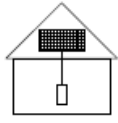
PHOTOVOLTAIC POINT OF INTERCONNECTION WARNING: ELECTRIC SHOCK HAZARD. DO NOT TOUCH TERMINALS. TERMINALS ON BOTH THE LINE AND LOAD SIDE MAY BE ENERGIZED IN THE OPEN POSITION. FOR SERVICE DE-ENERGIZE BOTH SOURCE AND MAIN BREAKER PV POWER SOURCE <table border="1"> <tr> <td>MAXIMUM RATED AC OUTPUT CURRENT:</td> <td>25.6</td> <td>AMPS</td> </tr> <tr> <td>MAXIMUM OPERATING AC VOLTAGE:</td> <td>240</td> <td>VOLTS</td> </tr> </table> SIGNAGE LOCATIONS: - MAIN SERVICE PANEL - NEW INDOOR / OUTDOOR LOAD CENTER	MAXIMUM RATED AC OUTPUT CURRENT:	25.6	AMPS	MAXIMUM OPERATING AC VOLTAGE:	240	VOLTS	RAPID SHUTDOWN SWITCH FOR SOLAR PV SYSTEM SIGNAGE LOCATIONS: - AC DISCONNECT - LABEL SHALL BE INSTALLED NO MORE THAN 3' FROM THE SWITCH
MAXIMUM RATED AC OUTPUT CURRENT:	25.6	AMPS					
MAXIMUM OPERATING AC VOLTAGE:	240	VOLTS					
PHOTOVOLTAIC SYSTEM AC DISCONNECT <table border="1"> <tr> <td>RATED AC OUTPUT CURRENT:</td> <td>25.6</td> <td>AMPS</td> </tr> <tr> <td>NOMINAL OPERATING AC VOLTAGE:</td> <td>240</td> <td>VOLTS</td> </tr> </table> SIGNAGE LOCATIONS: INDOOR / OUTDOOR AC DISCONNECT	RATED AC OUTPUT CURRENT:	25.6	AMPS	NOMINAL OPERATING AC VOLTAGE:	240	VOLTS	WARNING POWER SOURCE OUTPUT CONNECTION DO NOT RELOCATE THIS OVERCURRENT DEVICE SIGNAGE LOCATIONS: - MAIN SERVICE PANEL - NEW INDOOR / OUTDOOR LOAD CENTER - INDOOR / OUTDOOR SUB PANEL
RATED AC OUTPUT CURRENT:	25.6	AMPS					
NOMINAL OPERATING AC VOLTAGE:	240	VOLTS					
SIGNAGE NOTES 1. MATERIAL USED FOR THE SIGNAGE SHALL BE REFLECTIVE, WEATHER RESISTANT AND SUITABLE FOR THE ENVIRONMENT. 2. ALL SIGNAGE SHALL HAVE ALL CAPITAL LETTERS WITH MINIMUM 1/2" LETTER HEIGHT, WHITE ON RED BACKGROUND. 3. MAIN SERVICE DISCONNECT MARKING SHALL BE PLACED ADJACENT TO MAIN SERVICE DISCONNECT IN A LOCATION CLEARLY VISIBLE FROM THE LOCATION WHERE THE LEVER IS OPERATED. 4. MARKING IS REQUIRED ON ALL INTERIOR AND EXTERIOR DC CONDUIT, RACEWAYS, ENCLOSURES, CABLE ASSEMBLIES, AND JUNCTION BOXES TO ALERT THE FIRE SERVICE TO AVOID CUTTING THEM. MARKINGS SHALL BE PLACED EVERY 10', AT TURNS AND ABOVE AND/OR BELOW PENETRATIONS, AND AT ALL DC COMBINER AND JUNCTION BOXES. 5. DO NOT USE SCREWS FOR SIGNAGE ATTACHMENT. USE ONLY APPROVED ADHESIVE.	AC SOLAR DISCONNECT SIGNAGE LOCATIONS: INDOOR / OUTDOOR AC DISCONNECT						
ENERGY STORAGE SYSTEM CONTAINS ENERGIZED BATTERIES SIGNAGE LOCATIONS: ENPHASE ENERGY STORAGE SYSTEM	PHOTOVOLTAIC SYSTEM COMBINER PANEL: DO NOT ADD LOADS SIGNAGE LOCATIONS: NEW INDOOR / OUTDOOR DC COMBINER						
SOLAR PV BREAKER BREAKER IS BACKFED DO NOT RELOCATE SIGNAGE LOCATIONS: - MAIN SERVICE PANEL - PANEL ENCLOSURE EXTERIOR	SOLAR PV SYSTEM EQUIPPED WITH RAPID SHUTDOWN TURN RAPID SHUTDOWN SWITCH TO THE "OFF" POSITION TO SHUTDOWN PV SYSTEM AND REDUCE SHOCK HAZARD IN THE ARRAY  SIGNAGE LOCATIONS: SHALL BE INSTALLED NO MORE THAN 3' FROM THE SERVICE DISCONNECTING MEANS TO WHICH THE PV SYSTEMS ARE CONNECTED.						

FIGURE 1: ELECTRICAL SIGNAGES/STICKERS

WARNING TURN OFF UTILITY DG AC SAFETY DISCONNECT PRIOR TO WORKING IN METER SERVICE PANEL SIGNAGE LOCATIONS: - MAIN SERVICE PANEL - PANEL ENCLOSURE EXTERIOR	WARNING INVERTER OUTPUT CONNECTION DO NOT RELOCATE THIS OVERCURRENT DEVICE SIGNAGE LOCATIONS: - MAIN SERVICE PANEL - PANEL ENCLOSURE EXTERIOR
WARNING POWER SUPPLY SOURCES UTILITY GRID PV ELECTRICAL SYSTEM SIGNAGE LOCATIONS: - INDOOR / OUTDOOR AC DISCONNECT - MAIN SERVICE PANEL - PANEL ENCLOSURE EXTERIOR	UTILITY PV SYSTEM DISCONNECT SAFETY SWITCH FOR UTILITY OPERATION AC DISCONNECT SIGNAGE LOCATIONS: INDOOR / OUTDOOR AC DISCONNECT

FIGURE 2: WIRING ROOF PENETRATION CROSS SECTION

